

Eran David Tzarum

Software Engineer

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PROFILE

Computer Science graduate (B.Sc.) specializing in Backend Development and Linux environments. Proven experience architecting robust systems in Python, Java, and C/C++. Highly proficient in proactively utilizing AI-assisted development tools (Claude Code, Cursor) to accelerate development cycles, enhance code quality, and rapidly integrate new technologies. Passionate about open-source principles and building scalable, enterprise-ready infrastructure.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, SQL, C#, Dart, JavaScript
Infrastructure & DevOps: Linux/Unix, Git/GitHub, Docker, Kubernetes (Core concepts), CI/CD flows, Network Protocols (TCP/UDP), AWS basics, Microservices architecture
AI Dev Tools: Claude Code, Cursor, Windsurf, GitHub Copilot
Backend & Data: REST APIs, Supabase (PostgreSQL), Firebase, Google Gemini API, Scikit-learn

SOFTWARE ENGINEERING EXPERIENCE

- Communication Networks** | *Python, C, TCP/UDP, Linux Sockets* Independent Project
- TCP/UDP Programming (C):** Solved complex networking problems by developing reliable data transfer protocols and congestion control mechanisms using raw Linux OS sockets.
 - Custom Protocol Stack (Python):** Built a low-level network ecosystem from scratch, implementing a Reliable UDP (RUDP) protocol with a Go-Back-N sliding window and custom DHCP/DNS servers.
- Smart Fridge - Startup Project (8200 IMPACT)** | *Python, Java, Supabase, Gemini API* 2022 - 2023
- Engineered the core logic and database architecture for an MVP (**Java, Firebase**), implementing a custom ingredient-matching algorithm that improved recipe recommendation accuracy based on dynamic user inventory.
 - [Present]** Architecting a scalable backend infrastructure (**Python, Supabase**) integrating Google Gemini 2.5 Flash for OCR receipt ingestion, data processing, and conversational features.
- "TimeSkip" - AI-Based Assessment Platform** | *Python, REST APIs, PostgreSQL* Grade: 96
- Led the end-to-end backend development of a diagnostic system, tested on **104 users** in a real-world educational setting to evaluate behavioral patterns.
 - Engineered a robust **Python/REST API** backend to handle high-frequency event logging, integrated securely with Supabase (PostgREST) data storage.

ADDITIONAL EXPERIENCE

- "Tziyun LaShevah" Learning Center** Herzliya, Israel
Math Instructor & Operations Manager 2021 - Present
- Mentored high-school students in advanced mathematics (5 units) and managed daily operations, scheduling, and customer relations under high-responsibility conditions during owner's absence.

EDUCATION

- Ariel University** Ariel, Israel
B.Sc. in Computer Science 2026
- Academic Excellence:** Achieved a **96.6 average** in the final year.
 - Selected Coursework:** Operating Systems (100), Data Compression (98), Cyber Attack Detection (90).
- 8200 IMPACT & Shenkar** Tel Aviv, Israel
Impact-Tech Accelerator Program Alumni 2022 - 2023
- Graduated a highly selective acceleration program for impact-tech ventures tackling environmental challenges.

MILITARY SERVICE

- IDF Reserve Duty - "Iron Swords" War** Israel
Operational Duty (Volunteer) 2024 - 2025
- Maintained a **96.6 academic average** while serving over **200 days** in high-pressure operational reserve duty.